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Effect of M_xO ($M = \text{Li}, \text{K}, \text{Ca}$) on the solubility of Ag_2O in borate glass

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Abstract

The variation of Ag_2O solubility under air atmosphere with the chemical potential of Ag was measured for the $M_xO\text{--}B_2O_3\text{--}Al_2O_3$ system by equilibrating with Ag–Au alloy of various Ag contents at 1273 and 1373 K. It was found that in this system the regular solution model is not valid for estimating the activity coefficient over a wide composition range. The activity coefficient of Ag_2O in B_2O_3 at infinite dilution and first-order Gibbs energy interaction coefficients of Ag_2O , Al_2O_3 , and Na_2O on Ag_2O were determined from the Ag_2O solubility data without assuming a regular solution. The increase of the Al_2O_3 solubility by addition of alkali oxide suggested that the glass melts were equilibrated with $2Al_2O_3 \cdot 9B_2O_3$ and not Al_2O_3 and the activity was considered to increase by the addition of alkali oxide. A good correlation was found between the activity coefficient of Ag_2O and the optical basicity. © 2002 Elsevier Science B.V. All rights reserved.

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1. Introduction

Since silver is a noble element, its oxide is easily reduced at high temperature. The decomposition temperature of Ag_2O under air atmosphere is calculated to be 421 K from its thermodynamic data [1]. In glass melts, however, Ag_2O can exist in a stable form at a limited content even at high temperatures. Ag_2O reacts as a basic oxide, so that it becomes very stable in glass melts of acidic composition. Therefore, the production of Ag_2O

containing glass which has attracted interest for its antibacterial activity [2] is possible.

The solubility of Ag_2O in glass melts is an important factor for effective production of Ag_2O containing glass. Here, the solubility of Ag_2O is defined as the Ag_2O content in the glass melt equilibrated with Ag. The Ag_2O added to the glass batch over its solubility is reduced to metallic silver and is of no use to increase Ag_2O content in the glass. The solubility of Ag_2O depends on the oxygen potential and temperature during melting and the glass composition. The glass composition should be selected carefully because it is difficult to control the oxygen potential and temperature in the melting process. The stability of Ag_2O can be represented in terms of its activity coefficient,

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$\gamma_{\text{Ag}_2\text{O}}$. The variation of $\gamma_{\text{Ag}_2\text{O}}$ by the addition of glass components is a significant information necessary to select glass composition. The authors have evaluated the effect of Na_2O , Al_2O_3 and Ag_2O on $\gamma_{\text{Ag}_2\text{O}}$ in terms of the interaction parameter $\varepsilon_{\text{Ag}_2\text{O}}^i$ ($i = \text{Na}_2\text{O}$, Ag_2O , Al_2O_3) and showed that Ag_2O behaves as a basic oxide [3,4]. In this study, the effect of Li_2O , K_2O and CaO on $\gamma_{\text{Ag}_2\text{O}}$ or the solubility of Ag_2O was measured and discussed from the view point of basicity of the glass.

2. Experimental

Reagent grade B_2O_3 , Li_2CO_3 , K_2CO_3 , CaCO_3 and Al_2O_3 were used as starting materials. Well mixed batches of a given composition were pre-melted at 1473 K to obtain $\text{M}_x\text{O}-\text{B}_2\text{O}_3-\text{Al}_2\text{O}_3$ glasses. The Al_2O_3 content in this glass was adjusted so that it became close to Al_2O_3 saturation which was known from preliminary experiments. The obtained glass (2.5 g) was equilibrated with Ag (5 g) in an alumina crucible at 1273 or 1373 K for 24 h under atmospheric air as shown in Fig. 1. The glass melt was contacted with the inner sur-

face of the alumina crucible and the Ag–Au alloy was exposed to both the glass melt and air. Preliminary experiments showed that the equilibrium between the glass melt and silver and that between the glass melt and alumina crucible were attained in 24 h.

After the equilibration, the sample was quenched and the glass composition was analyzed. Volhard method [5] for Ag_2O , flame spectrochemical method [6] for Li_2O and K_2O , KMnO_4 titration method for CaO and neutralization titration method [6] for B_2O_3 were employed for chemical analysis.

3. Results

The variations of Ag_2O content in mole fraction ($X_{\text{Ag}_2\text{O}}$) in $\text{M}_x\text{O}-\text{B}_2\text{O}_3-\text{Al}_2\text{O}_3$ glasses with M_xO content at 1273 and 1373 K were shown in Figs. 2 and 3, respectively, and Table 1. The data for $\text{Na}_2\text{O}-\text{B}_2\text{O}_3-\text{Al}_2\text{O}_3$ glasses reported in the previous report [3] were shown together in the figures. The error in the compositional analysis is ± 0.001 . Ag_2O content decreased with the increase of M_xO

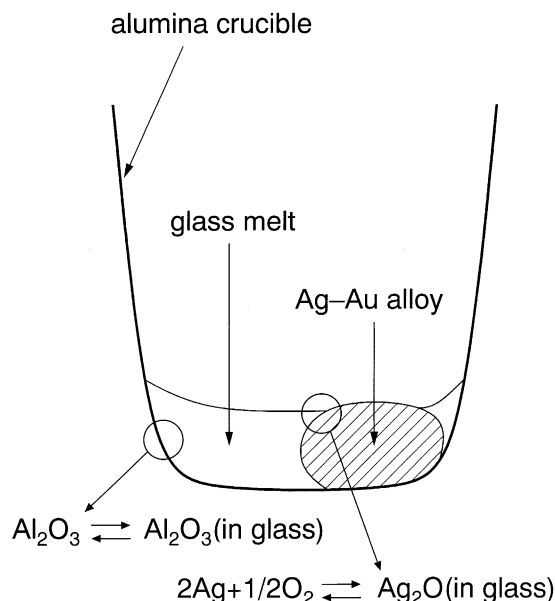


Fig. 1. Schematic drawing of glass melting in contact with Al_2O_3 (alumina crucible) and Ag–Au alloy exposed to air.

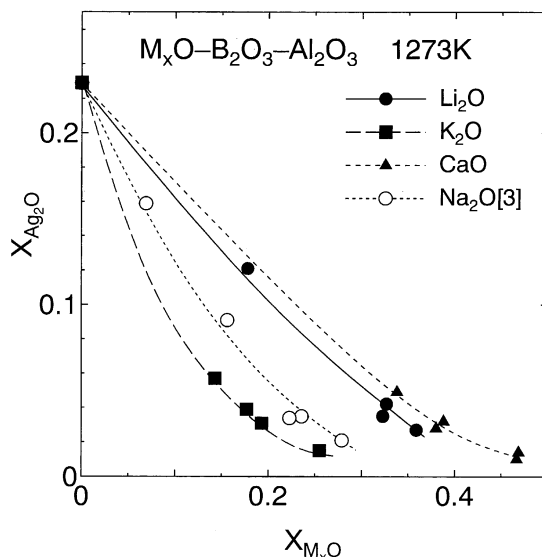


Fig. 2. Variation of Ag_2O content with M_xO content in $\text{M}_x\text{O}-\text{B}_2\text{O}_3-\text{Al}_2\text{O}_3$ glass at 1273 K. The line is a guide to the eye.

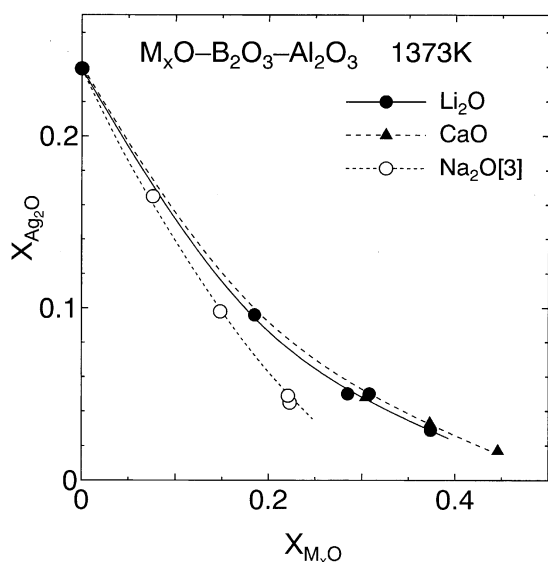
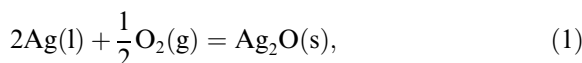


Fig. 3. Variation of Ag_2O content with M_xO content in $\text{M}_x\text{O}-\text{B}_2\text{O}_3-\text{Al}_2\text{O}_3$ glass at 1373 K. The line is a guide to the eye.

content and finally it became one-tenth or less. On the other hand, the effect of temperature on Ag_2O content is not significant.

The order of Ag_2O content at the same M_xO content is $\text{CaO} \approx \text{Li}_2\text{O} > \text{Na}_2\text{O} > \text{K}_2\text{O}$ though Al_2O_3 content is different in each glass. This is just the opposite order of the basicity of oxide, M_xO . The effect of the basicity on Ag_2O content will be discussed in the following section.

The activity coefficient of Ag_2O was calculated by considering the following reaction, where l, g, and s in parentheses denote the reference states



$$\Delta G_1^0 = -53,100 + 84.1T \quad (\text{J/mol}) \quad (2)$$

(cf. [1]), and

$$K_1 = \exp\left(-\frac{\Delta G_1^0}{RT}\right) = \frac{a_{\text{Ag}_2\text{O}}}{a_{\text{Ag}}^2 P_{\text{O}_2}^{1/2}} = \frac{\gamma_{\text{Ag}_2\text{O}} X_{\text{Ag}_2\text{O}}}{a_{\text{Ag}}^2 P_{\text{O}_2}^{1/2}}, \quad (3)$$

where ΔG_1^0 and K_1 are the standard free energy change and the equilibrium constant of Eq. (1), respectively, and R is the gas constant, T is the temperature, a_i and γ_i are activity and activity coefficient of the component ' i ', respectively, and

Table 1

Glass composition after equilibration with Ag, activity coefficient of Ag_2O and A_{th}

No.	Temperature (K)	$X_{\text{Ag}_2\text{O}}$	$X_{\text{B}_2\text{O}_3}$	$X_{\text{Li}_2\text{O}}$	$X_{\text{K}_2\text{O}}$	X_{CaO}	$X_{\text{Al}_2\text{O}_3}$	$\gamma_{\text{Ag}_2\text{O}}$	A_{th}
11	1273	0.121	0.468	0.178	—	—	0.233	−3.803	0.520
12	1273	0.035	0.389	0.323	—	—	0.253	−2.562	0.566
13	1273	0.042	0.364	0.327	—	—	0.267	−2.745	0.572
14	1273	0.027	0.342	0.359	—	—	0.272	−2.303	0.584
15	1373	0.096	0.474	0.185	—	—	0.245	−3.936	0.523
16	1373	0.050	0.397	0.285	—	—	0.268	−3.284	0.558
17	1373	0.050	0.395	0.308	—	—	0.247	−3.284	0.561
18	1373	0.029	0.344	0.374	—	—	0.253	−2.739	0.585
41	1273	0.057	0.619	—	0.143	—	0.181	−3.050	0.514
42	1273	0.039	0.554	—	0.177	—	0.230	−2.671	0.538
43	1273	0.031	0.507	—	0.193	—	0.269	−2.441	0.553
44	1273	0.015	0.396	—	0.255	—	0.334	−1.715	0.596
51	1273	0.049	0.459	—	—	0.338	0.154	−2.899	0.549
52	1273	0.032	0.285	—	—	0.388	0.295	−2.473	0.602
53	1273	0.028	0.291	—	—	0.380	0.301	−2.339	0.600
54	1273	0.014	0.238	—	—	0.469	0.279	−1.646	0.632
55	1273	0.010	0.315	—	—	0.467	0.208	−1.310	0.611
56	1373	0.048	0.424	—	—	0.304	0.224	−3.244	0.554
57	1373	0.033	0.387	—	—	0.373	0.207	−2.869	0.574
58	1373	0.017	0.337	—	—	0.446	0.200	−2.206	0.601
59	1373	0.012	0.314	—	—	0.502	0.172	−1.857	0.618

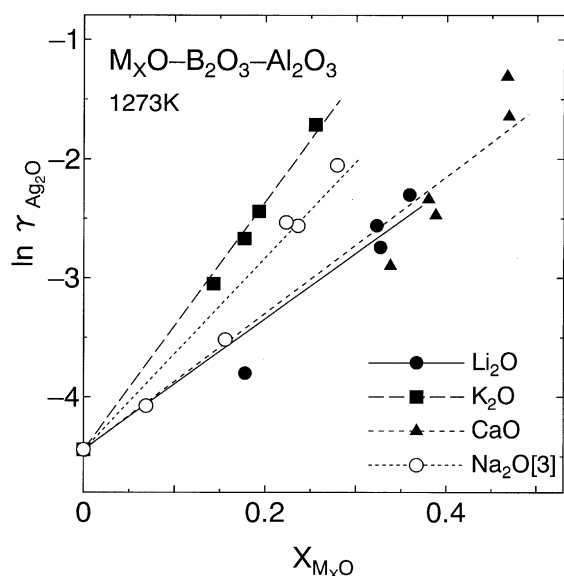


Fig. 4. Variation of $\gamma_{\text{Ag}_2\text{O}}$ with M_xO content in $\text{M}_x\text{O}-\text{B}_2\text{O}_3-\text{Al}_2\text{O}_3$ glass at 1273 K. The lines are a guide to the eye.

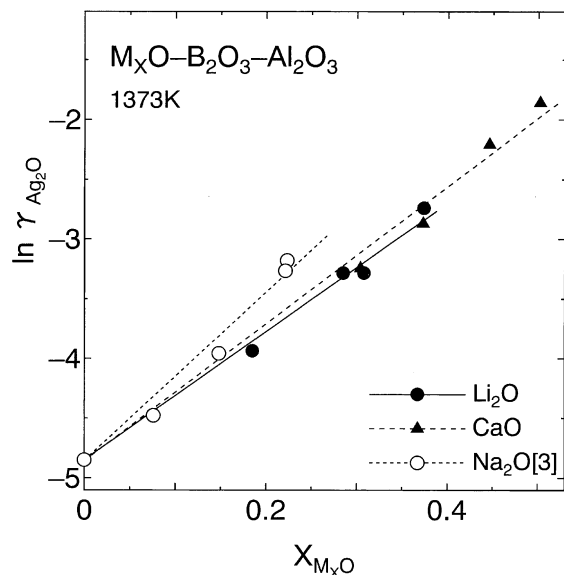


Fig. 5. Variation of $\gamma_{\text{Ag}_2\text{O}}$ with M_xO content in $\text{M}_x\text{O}-\text{B}_2\text{O}_3-\text{Al}_2\text{O}_3$ glass at 1373 K. The lines are a guide to the eye.

P_{O_2} oxygen partial pressure under air (0.21 atm). Solid state was chosen as the reference state of Ag_2O because it decomposes at the experimental temperature without melting [7].

Calculated $\gamma_{\text{Ag}_2\text{O}}$ for three systems are shown in Figs. 4 and 5 together with that for the $\text{Na}_2\text{O}-\text{B}_2\text{O}_3-\text{Al}_2\text{O}_3$ system. It is found that $\gamma_{\text{Ag}_2\text{O}}$ increases linearly with the increase of M_xO , and decreases as temperature increases.

4. Discussion

4.1. Evaluation of $\varepsilon_{\text{Ag}_2\text{O}}^{\text{M}_x\text{O}}$

The $\gamma_{\text{Ag}_2\text{O}}$ in the study is expressed as a function of composition by using the first-order interaction parameters, ε as follows [8]:

$$\ln \gamma_{\text{Ag}_2\text{O}} = \ln \gamma_{\text{Ag}_2\text{O}}^0 + \varepsilon_{\text{Ag}_2\text{O}}^{\text{M}_x\text{O}} X_{\text{M}_x\text{O}} + \varepsilon_{\text{Ag}_2\text{O}}^{\text{Al}_2\text{O}_3} X_{\text{Al}_2\text{O}_3} + \varepsilon_{\text{Ag}_2\text{O}}^{\text{Ag}_2\text{O}} X_{\text{Ag}_2\text{O}}, \quad (4)$$

where $\gamma_{\text{Ag}_2\text{O}}^0$ and $\varepsilon_{\text{Ag}_2\text{O}}^i$ are the activity coefficient of Ag_2O in B_2O_3 and the first-order interaction parameter of the i th component on Ag_2O , respectively.

When $\ln \gamma_{\text{Ag}_2\text{O}} - \varepsilon_{\text{Ag}_2\text{O}}^{\text{Al}_2\text{O}_3} X_{\text{Al}_2\text{O}_3} - \varepsilon_{\text{Ag}_2\text{O}}^{\text{Ag}_2\text{O}} X_{\text{Ag}_2\text{O}}$ is plotted against $X_{\text{M}_x\text{O}}$, $\varepsilon_{\text{Ag}_2\text{O}}^{\text{M}_x\text{O}}$ is obtained from the slope calculated by linear regression. In the previous report, $\gamma_{\text{Ag}_2\text{O}}^0$, $\varepsilon_{\text{Ag}_2\text{O}}^{\text{Ag}_2\text{O}}$ and $\varepsilon_{\text{Ag}_2\text{O}}^{\text{Al}_2\text{O}_3}$ were reported to be -6.7, 6.9 and -0.6 at 1373 K, respectively. Those values at 1273 K were evaluated as follows.

The temperature dependence of the thermodynamic parameters was often estimated by the regular solution model. Activity coefficient is related to excess free energy, $\Delta \bar{G}^E$ according to Eq. (5):

$$RT \ln \gamma = \Delta \bar{G}^E = \Delta \bar{H} - T \Delta \bar{S}^E. \quad (5)$$

Assuming a regular solution, $\Delta \bar{S}^E$ is equal to zero and $RT \ln \gamma$ becomes constant independent of temperature. According to Eq. (4), this means that $\ln \gamma_{\text{Ag}_2\text{O}}^0$ and $\varepsilon_{\text{Ag}_2\text{O}}^i$ are in inverse proportion to temperature. However, it was reported that the variation of $\gamma_{\text{Ag}_2\text{O}}$ with Ag_2O content does not obey the regular solution model [4] and the temperature dependence of $\gamma_{\text{Ag}_2\text{O}}$ shown in Figs. 4 and 5 contradicts the model. Since it is not reasonable to evaluate the temperature dependence of $\varepsilon_{\text{Ag}_2\text{O}}^{\text{Ag}_2\text{O}}$ and $\varepsilon_{\text{Ag}_2\text{O}}^{\text{Al}_2\text{O}_3}$ based on the regular solution model, the values at 1373 K were used at 1273 K.

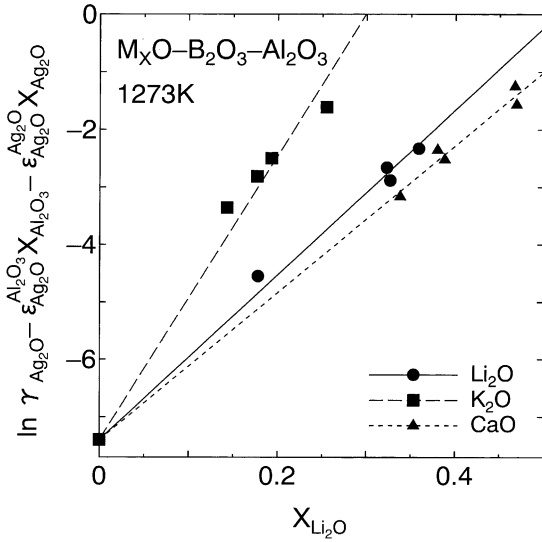


Fig. 6. Variation of $\ln \gamma_{\text{Ag}_2\text{O}} - \epsilon_{\text{Ag}_2\text{O}}^{\text{Al}_2\text{O}_3} X_{\text{Al}_2\text{O}_3} - \epsilon_{\text{Ag}_2\text{O}}^{\text{Ag}_2\text{O}} X_{\text{Ag}_2\text{O}}$ with $M_x\text{O}$ content in $M_x\text{O}-\text{B}_2\text{O}_3-\text{Al}_2\text{O}_3$ glass at 1273 K. The lines were obtained by linear regression.

$\ln \gamma_{\text{Ag}_2\text{O}} - \epsilon_{\text{Ag}_2\text{O}}^{\text{Al}_2\text{O}_3} X_{\text{Al}_2\text{O}_3} - \epsilon_{\text{Ag}_2\text{O}}^{\text{Ag}_2\text{O}} X_{\text{Ag}_2\text{O}}$ is plotted against $X_{M_x\text{O}}$ in Figs. 6 and 7. A good linear relationship was obtained in every figure. The line in

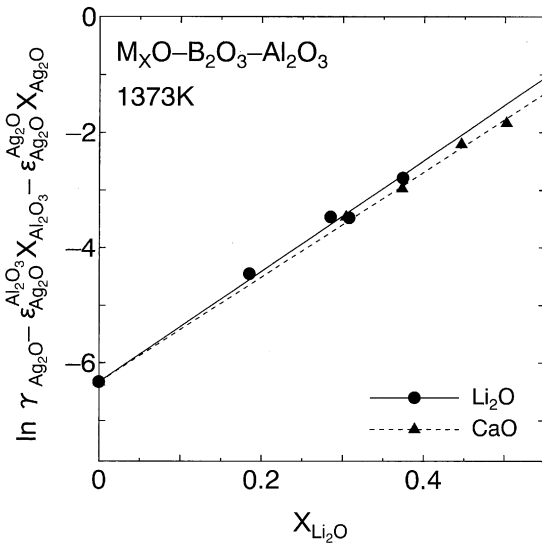


Fig. 7. Variation of $\ln \gamma_{\text{Ag}_2\text{O}} - \epsilon_{\text{Ag}_2\text{O}}^{\text{Al}_2\text{O}_3} X_{\text{Al}_2\text{O}_3} - \epsilon_{\text{Ag}_2\text{O}}^{\text{Ag}_2\text{O}} X_{\text{Ag}_2\text{O}}$ with $M_x\text{O}$ content in $M_x\text{O}-\text{B}_2\text{O}_3-\text{Al}_2\text{O}_3$ glass at 1373 K. The line was obtained by linear regression.

the figures was calculated by the least squares method. The interaction parameters calculated from the slope of the lines are shown in Table 2. The order of the interaction parameter is $\epsilon_{\text{Ag}_2\text{O}}^{\text{Ag}_2\text{O}} < \epsilon_{\text{Ag}_2\text{O}}^{\text{Li}_2\text{O}} < \epsilon_{\text{Ag}_2\text{O}}^{\text{CaO}} < \epsilon_{\text{Ag}_2\text{O}}^{\text{K}_2\text{O}}$. In view of the electronegativity, the element with small a electronegativity has a large interaction parameter with Ag_2O . Since the effect of alkali oxide on various properties of glasses is similar, the use of Li_2O is suitable for the production of Ag containing glass. For example, the production of Ag_2O containing glass with $X_{\text{Ag}_2\text{O}} = 0.1$ at 1273 K is compared for $\text{Li}_2\text{O}-\text{B}_2\text{O}_3$ and $\text{K}_2\text{O}-\text{B}_2\text{O}_3$ glasses. The maximum $X_{M_x\text{O}}$ are calculated to be 0.19 for Li_2O and 0.11 for K_2O . Then, the usage of Li_2O instead of K_2O has an advantage because the choice of glass composition extends.

4.2. Evaluation of $\epsilon_{\text{Al}_2\text{O}_3}^{M_x\text{O}}$

In this study, an Al_2O_3 crucible was employed and no reaction product was observed at the interface between the glass melt and the crucible. Therefore, the melt was saturated with Al_2O_3 , or the activity of Al_2O_3 was unity. Then the activity coefficient of Al_2O_3 is expressed as follows:

$$\gamma_{\text{Al}_2\text{O}_3} = \frac{1}{X_{\text{Al}_2\text{O}_3}}. \quad (6)$$

On the other hand, $\gamma_{\text{Al}_2\text{O}_3}$ is written in terms of the interaction parameters as follows [8]:

$$\ln \gamma_{\text{Al}_2\text{O}_3} = \ln \gamma_{\text{Al}_2\text{O}_3}^0 + \epsilon_{\text{Al}_2\text{O}_3}^{\text{Al}_2\text{O}_3} X_{\text{Al}_2\text{O}_3} + \epsilon_{\text{Al}_2\text{O}_3}^{M_x\text{O}} X_{M_x\text{O}} + \epsilon_{\text{Al}_2\text{O}_3}^{\text{Ag}_2\text{O}} X_{\text{Ag}_2\text{O}}. \quad (7)$$

Theoretically, it has been shown that $\epsilon_j^i = \epsilon_i^j$ [8]. The value of $\epsilon_{\text{Al}_2\text{O}_3}^{\text{Al}_2\text{O}_3}$ is not known but it is expected to be smaller than $\epsilon_{\text{Ag}_2\text{O}}^{\text{Al}_2\text{O}_3}$ because Al_2O_3 is an amphoteric oxide. If the second term in Eq. (7) is neglected, the slope of the line obtained from the plot of $\ln \gamma_{\text{Ag}_2\text{O}} - \epsilon_{\text{Ag}_2\text{O}}^{\text{Al}_2\text{O}_3} X_{\text{Al}_2\text{O}_3}$ against $X_{M_x\text{O}}$ gives $\epsilon_{\text{Al}_2\text{O}_3}^{M_x\text{O}}$. Fig. 8 shows this plot for $M_x\text{O}-\text{B}_2\text{O}_3-\text{Al}_2\text{O}_3$ system ($M = \text{Li}, \text{Na}, \text{K}$ and Ca). Although the scattering of data is somewhat large, it is found that the variation of $\ln \gamma_{\text{Ag}_2\text{O}} - \epsilon_{\text{Ag}_2\text{O}}^{\text{Al}_2\text{O}_3} X_{\text{Al}_2\text{O}_3}$ with $M_x\text{O}$ content is not significant. From the slope of the line calculated by the least squares method, $\epsilon_{\text{Al}_2\text{O}_3}^{M_x\text{O}}$ was calculated between -0.6 and 0.2 . The

Table 2

Interaction parameters of M_xO on Ag_2O and $\epsilon_{Ag_2O}^{M_xO}$

	$\epsilon_{Ag_2O}^{Li_2O}$	$\epsilon_{Ag_2O}^{K_2O}$	$\epsilon_{Ag_2O}^{CaO}$	$\epsilon_{Ag_2O}^{Na_2O}$	$\epsilon_{Ag_2O}^{Ag_2O}$	$\epsilon_{Ag_2O}^{Al_2O_3}$
1273 K	10.2	17.3	9.6		7.4	−0.6
1373 K	9.4	—	9.0	15.6 ± 0.6	6.9 ± 0.7	$−0.6 \pm 0.6$

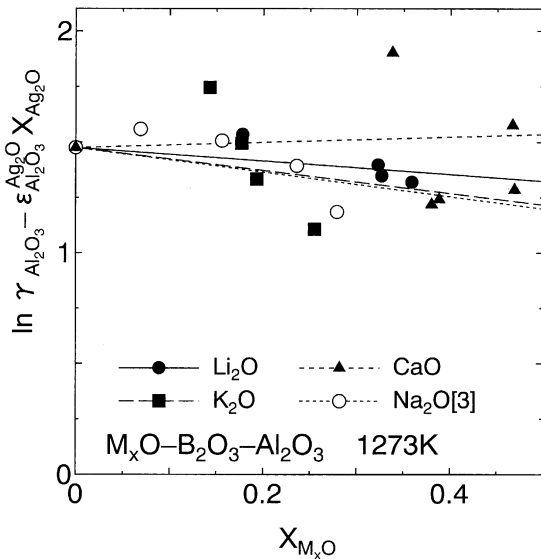


Fig. 8. Variation of $\ln \gamma_{Al_2O_3} - \epsilon_{Al_2O_3}^{Ag_2O} X_{Ag_2O}$ with M_xO content in $M_xO-B_2O_3-Al_2O_3$ glass. The lines were obtained by linear regression.

accuracy of this value is not good, but its magnitude is as small as $\epsilon_{Ag_2O}^{Al_2O_3}$ and its effect is negligible.

This result seems to contradict the fact that the solubility of Al_2O_3 for borate melts increases significantly by the addition of alkali oxide. For example, the solubility of Al_2O_3 at 1273 K is $X_{Al_2O_3} = 0.01$ in the $B_2O_3-Al_2O_3$ system and $X_{Al_2O_3} = 0.52$ in the $Li_2O-B_2O_3-Al_2O_3$ system when $X_{Li_2O} = 0.2$ [9]. In the $B_2O_3-Al_2O_3$ system, the solid phase equilibrated with the liquid phase is $2B_2O_3 \cdot 9Al_2O_3$ and not Al_2O_3 at 1273 K. Then, the activity of Al_2O_3 is not unity and changes with

composition in this case, which causes the increase of the solubility of Al_2O_3 . In the present experiments, the equilibrium phase with the melts was considered to be Al_2O_3 because Ag_2O or alkaline oxides content was high.

4.3. Relationship between γ_{Ag_2O} and optical basicity

Duffy and Ingram [10] proposed optical basicity as a measure of the basicity of oxide systems. The spectroscopic shifts observed in the $^1S_0 \rightarrow ^3P_1$ bands of Tl^+ , Pb^{2+} and Bi^{3+} were used to set up scales of basicity [10]. Based on this concept, theoretical optical basicity, A_{th} , is calculated according to Eq. (8):

$$A_{th} = \sum_i X_i A_i, \quad (8)$$

where X_i is the equivalent fraction based on the amount of oxygen i contributes to the overall glass stoichiometry and A_i is the optical basicity assigned to i . However, the optical basicity of Ag_2O has not been reported.

The optical basicity of oxides is expressed by the inverse of the basicity moderating parameter, γ , and it is reported that the following relationship is established between the basicity moderating parameter and the Pauling electronegativity:

$$\gamma = 1.36(x - 0.26). \quad (9)$$

The optical basicity of oxides in the present glasses was calculated by using the data in Table 3 and is shown in Table 1. The relationship between the optical basicity calculated by Eq. (8) and the activity coefficient of Ag_2O at 1373 K is shown in

Table 3

Theoretical optical basicity for various oxides

	Ag_2O	B_2O_3	Li_2O	K_2O	CaO	Al_2O_3
A_{th}	0.441	0.424	1.00	1.37	1.00	0.606

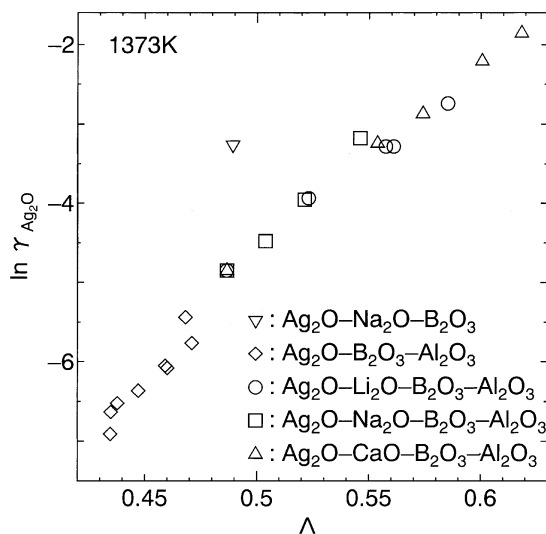


Fig. 9. Relationship between $\ln \gamma_{\text{Ag}_2\text{O}}$ and A_{th} .

Fig. 9. It is found that there is a good linear relationship between the optical basicity and activity coefficient of Ag_2O . However, the point for the $\text{Ag}_2\text{O-Na}_2\text{O-B}_2\text{O}_3$ glass is located far from the line. This sample contained no Al_2O_3 . This fact would mean that the optical basicity assigned for Al_2O_3 was not appropriate. $X_{\text{Al}_2\text{O}_3}$ in present glasses was about 0.2–0.3 and the effect of Al_2O_3 was the same except for $\text{Ag}_2\text{O-Na}_2\text{O-B}_2\text{O}_3$ glass. Then the relative basicity among the samples containing Al_2O_3 would become correct even if the optical basicity value assigned for Al_2O_3 were wrong. However, the error in their optical basicity becomes clear by comparing them with the glass containing no Al_2O_3 . This would be the reason the point for $\text{Ag}_2\text{O-Na}_2\text{O-B}_2\text{O}_3$ glass being located far from the line.

The optical basicity of Al_2O_3 calculated from the Pauling electronegativity is 1.65 which is more basic than B_2O_3 whose electronegativity is 2.27 [11]. The fact that $\epsilon_{\text{Ag}_2\text{O}}^{\text{Al}_2\text{O}_3}$ is close to zero suggests the basicity of Al_2O_3 is close to that of B_2O_3 because B_2O_3 is a solvent. Al_2O_3 is an amphoteric oxide and its basicity depends on the circumstance. Therefore, it is difficult to define the optical basicity of Al_2O_3 without considering glass composition.

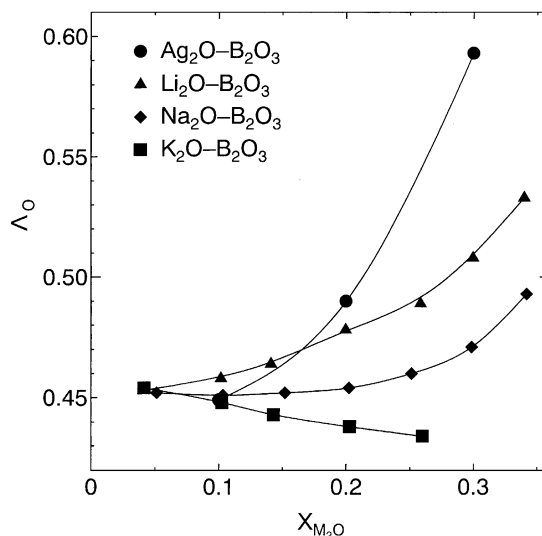


Fig. 10. Variation of A_0 with composition in $\text{M}_x\text{O-B}_2\text{O}_3$ glass [13]. The lines are guide to the eye.

Duffy offered an alternative definition of optical basicity, A_0 calculated from the polarizability of oxide ion, $\alpha_{\text{O}^{2-}}$, and showed following relationship [12]:

$$A_0 = 1.67(1 - 1/\alpha_{\text{O}^{2-}}). \quad (10)$$

Dimitrov and Komatsu [13] calculated A_0 for numerous binary glasses, and the variation of A_0 with composition in $\text{M}_2\text{O-B}_2\text{O}_3$ glass ($\text{M} = \text{Li}, \text{Na}, \text{K}, \text{Ag}$) is shown in Fig. 10. The variation of A_0 with composition is not large and A_0 of $\text{Ag}_2\text{O-B}_2\text{O}_3$ is maximum among four systems. Then this value does not express the basicity well.

5. Conclusion

The effect of Li_2O , K_2O , CaO and Al_2O_3 on the solubility of Ag_2O into aluminoborate glass was evaluated in terms of the interaction parameter, $\epsilon_{\text{Ag}_2\text{O}}^i$ ($i = \text{Li}_2\text{O}, \text{K}_2\text{O}, \text{CaO}$ and Al_2O_3). Their values are 10.2 for Li_2O , 17.3 for K_2O and 9.6 for CaO . $\epsilon_{\text{Al}_2\text{O}_3}^{\text{M}_x\text{O}}$ was close to zero and its effect is negligible. The increase of the solubility of Al_2O_3 by the addition of alkaline oxides was explained by the change of the equilibrium phase with melts from $2\text{B}_2\text{O}_3 \cdot 9\text{Al}_2\text{O}_3$ to Al_2O_3 . The order of interaction parameters between Ag_2O and M_xO is

reverse of that of electronegativity. It was found that Li_2O is a favorable component among alkali oxides for the production of silver containing glass. The activity coefficient has a good linear relationship with the optical basicity.

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